

PROBLEM 15.20 Working drawing (in.)

Assembly Name: Hydraulic Jack

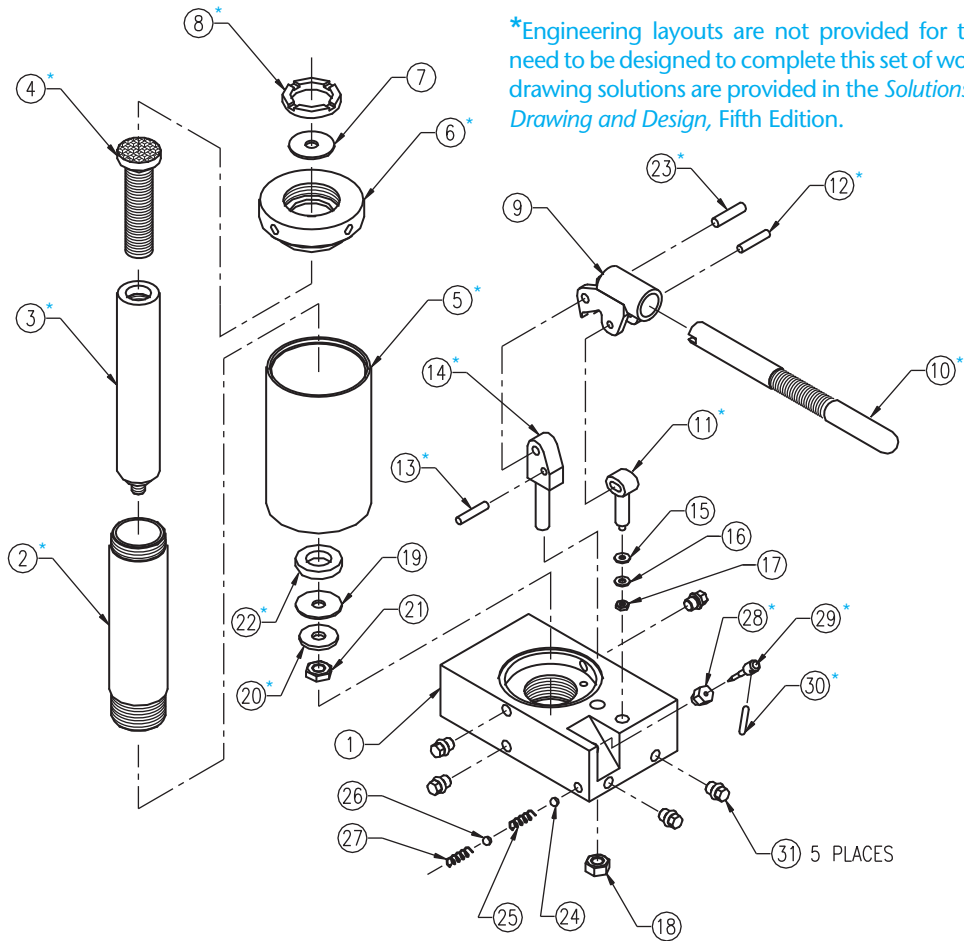
SPECIFIC INSTRUCTIONS: This problem is challenging and advanced.

Prepare a complete set of working drawings with one detail drawing per sheet and the assembly and parts list on another sheet. When preparing the assembly drawing, use separate

balloons for each part in each view or use only one balloon in the view that most clearly identifies the part. This is an engineering prototype project. Errors are likely in this problem. Verify dimensions during assembly. Establish tolerances between mating parts. Several purchase parts and parts need to be designed in this assembly. Manufacturer research is needed to locate available products.

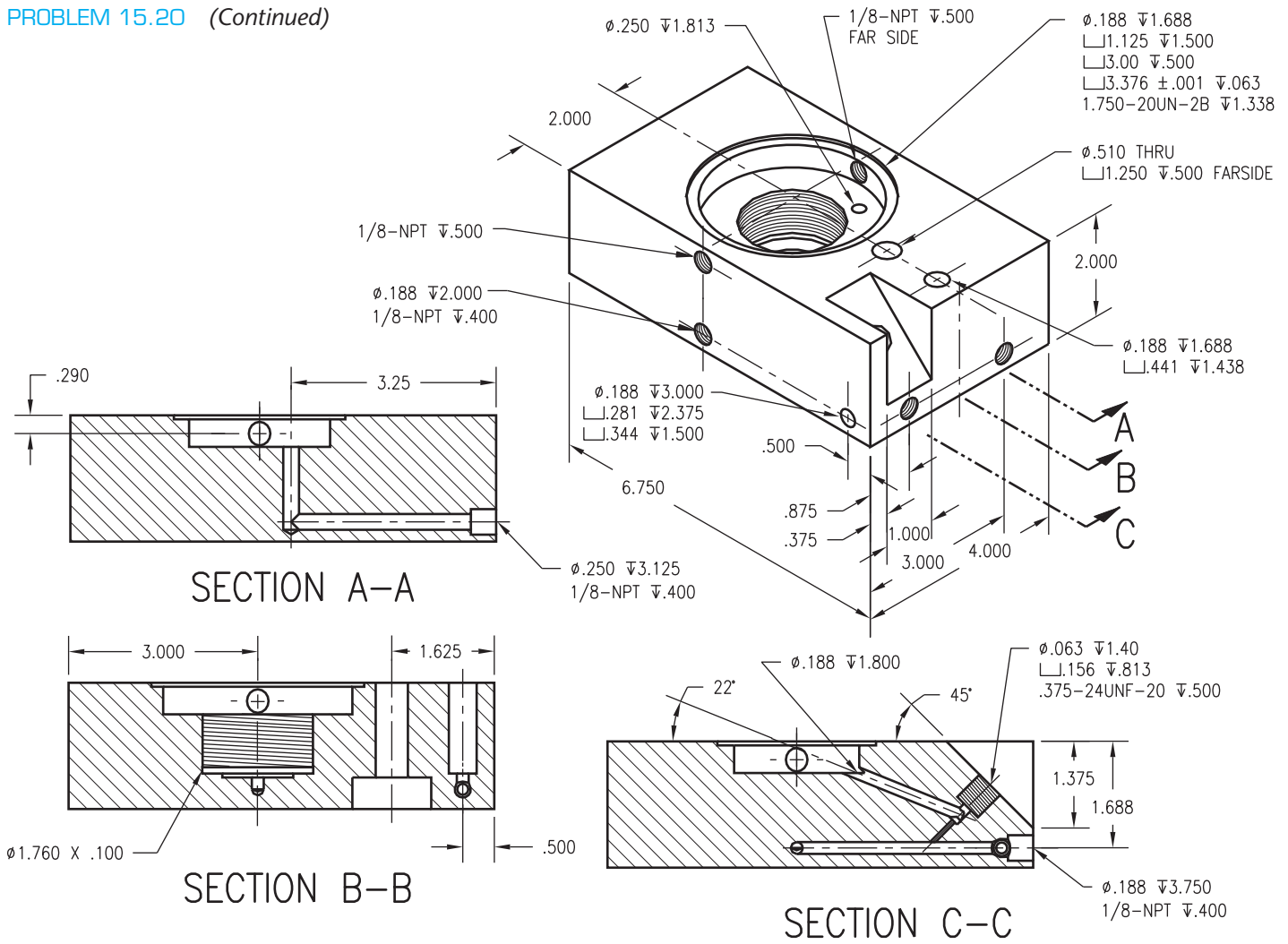
PARTS LIST

KEY	QTY	NAME	DESCRIPTION	PART NO.	KEY	QTY	NAME	DESCRIPTION	PART NO.
1	1	BASE	JACK	1DT1001	18	1	NUT	.437-14 UNC-2B	5DT1018
2*	1	TUBE	CYLINDER	1DT1002	19	1	CUP	Ø1.500 LEATHER	5DT1019
3*	1	PISTON	JACK	1DT1003	20*	1	WASHER	PISTON	1DT1020
4*	1	SCREW	JACK	1DT1004	21	1	NUT	.437-20 UNF-2B	5DT1021
5*	1	TUBE	RESERVOIR	1DT1005	22*	1	GUIDE	BRONZE	1DT1022
6*	1	CAP	TOP	1DT1006	23*	1	PIN	PIVOT	1DT1023
7	1	PACKING	Ø1.500 LEATHER	5DT1007	24	1	BALL	Ø.250	5DT1024
8*	1	NUT	PACKING	1DT1008	25	1	SPRING	Ø.281 × 1.000 CLOSED ENDS, 6 COILS	5DT1025
9	1	SOCKET	PUMP HANDLE	1DT1009	26	1	BALL	Ø.3125	5DT1026
10*	1	HANDLE	PUMP	1DT1010	27	1	SPRING	Ø.343 × 1.250 CLOSED ENDS, 9 COILS	5DT1027
11*	1	PLUNGER	PUMP	1DT1011	28*	1	NUT	NEEDLE VALVE	1DT1028
12*	1	PIN	DRIVE	1DT1012	29*	1	VALVE	NEEDLE	1DT1029
13*	1	PIN	STOP	1DT1013	30*	1	HANDLE	NEEDLE VALVE	1DT1030
14*	1	SUPPORT	PUMP PIVOT	1DT1014	31	5	PLUG	1/8-27 NPT	5DT1031
15	1	CUP SEAL	Ø.445 NEOPRENE	5DT1015					
16	1	WASHER	ANS TYPE B PLAIN NO. 10 N SERIES	5DT1016					
17	1	NUT	10-32 UNF-2B	5DT1017					

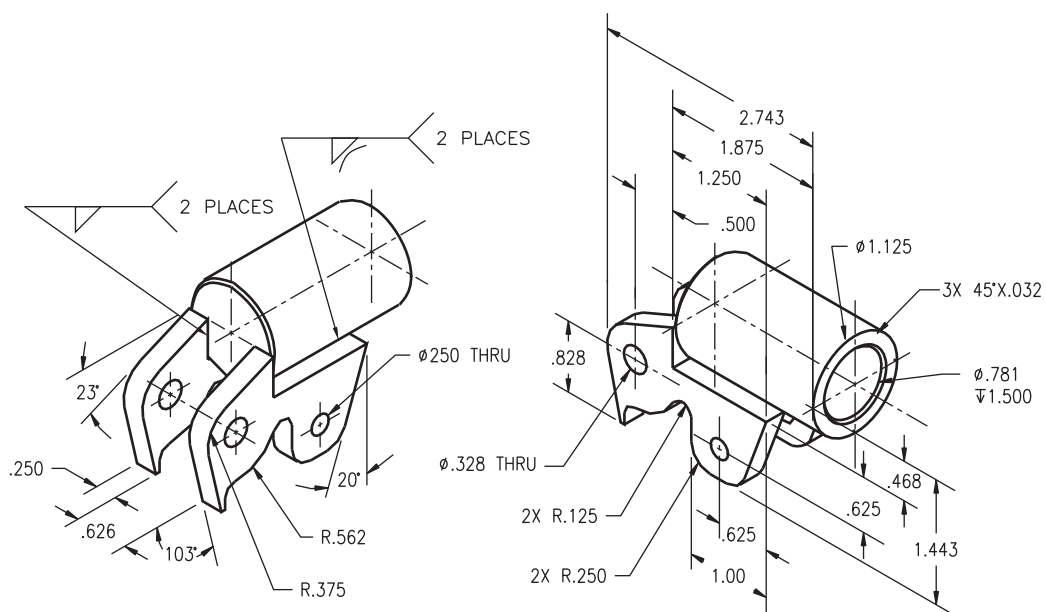


*Engineering layouts are not provided for these parts. These parts need to be designed to complete this set of working drawings. Possible drawing solutions are provided in the *Solutions Manual, for Engineering Drawing and Design, Fifth Edition*.

PROBLEM 15.20 (Continued)



① BASE

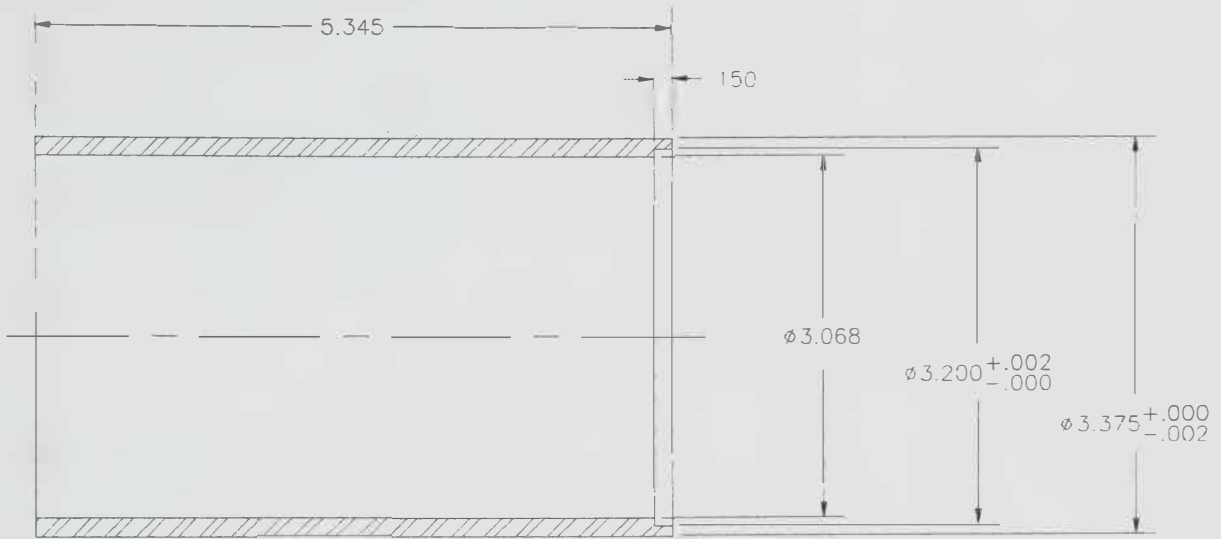


⑨ SOCKET

PROBLEM 18.20 Hydraulic jack (Continued)

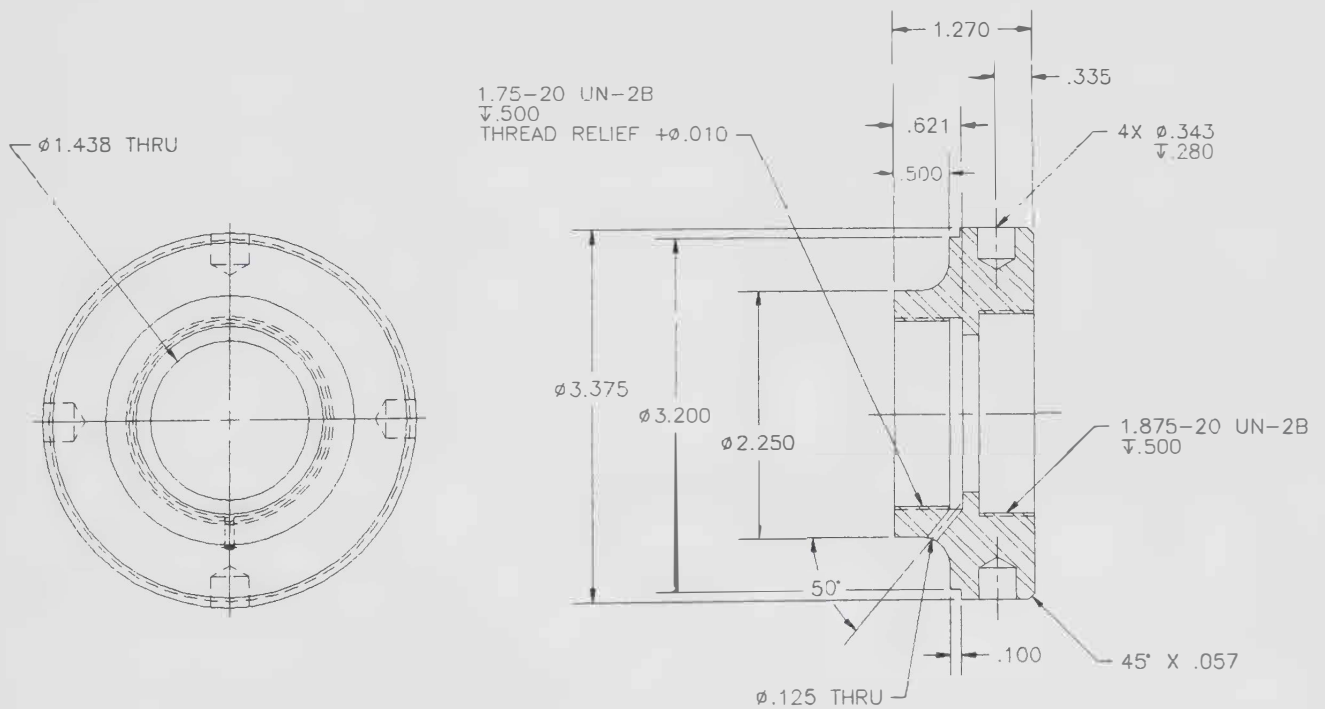
Part Name: Reservoir tube

Part No.: 1DT1005



Part Name: Top cap

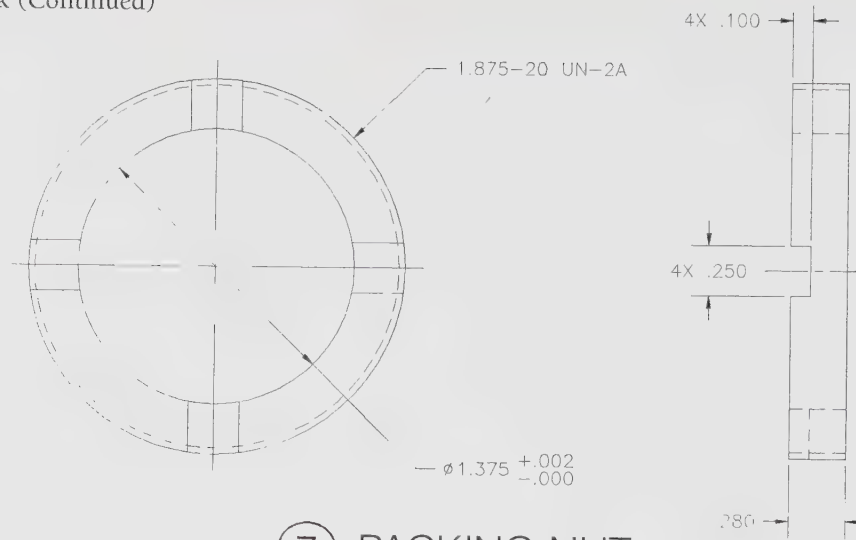
Part No.: 1DT1006



PROBLEM 18.20 Hydraulic jack (Continued)

Part Name: Packing nut

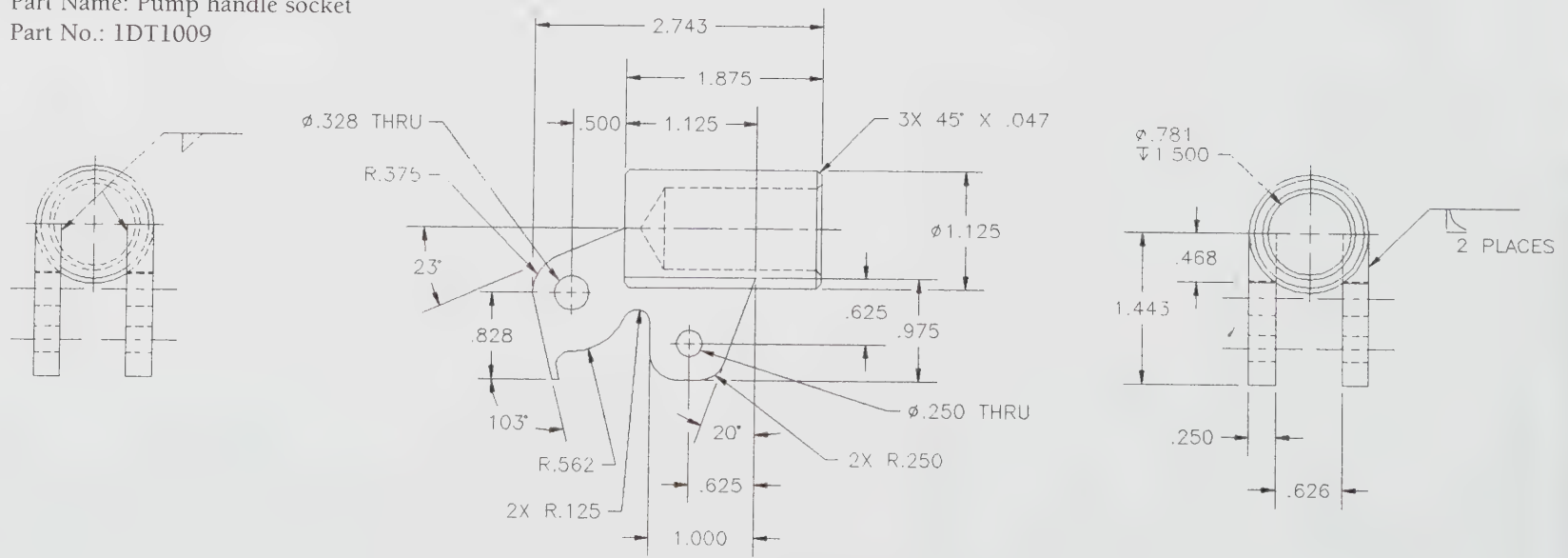
Part No.: 1DT1007



⑦ PACKING NUT

Part Name: Pump handle socket

Part No.: 1DT1009

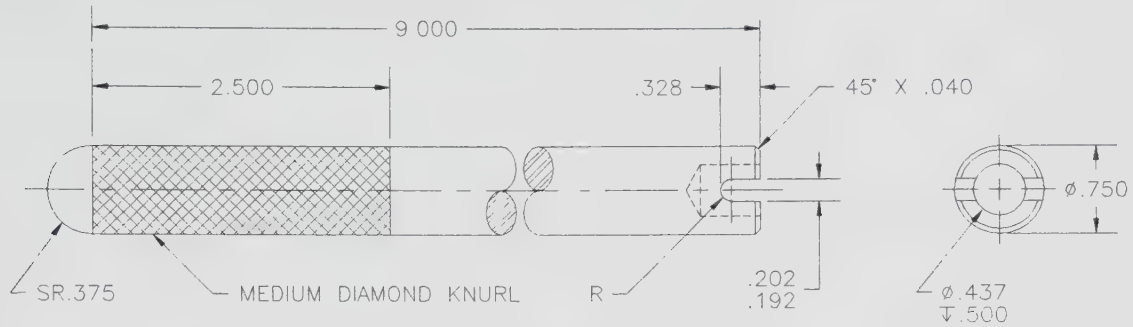


⑨ PUMP HANDLE SOCKET

PROBLEM 18.20 Hydraulic jack (Continued)

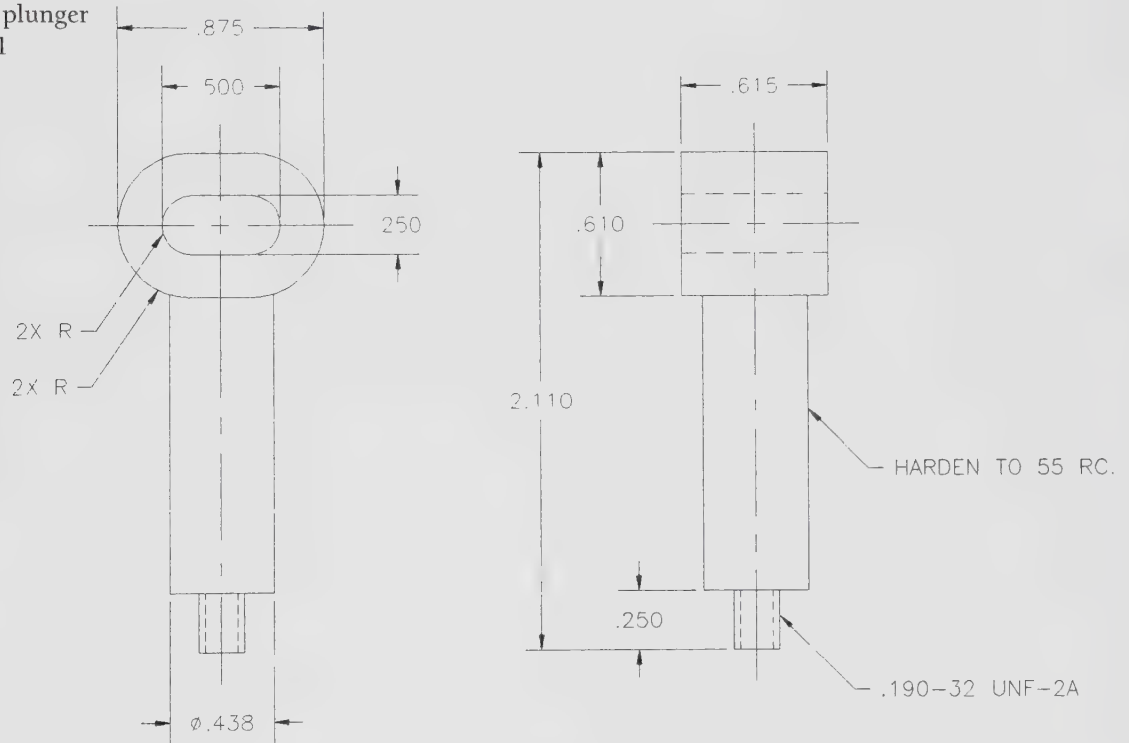
Part Name: Pump handle

Part No.: 1DT1010

**⑩ PUMP HANDLE**

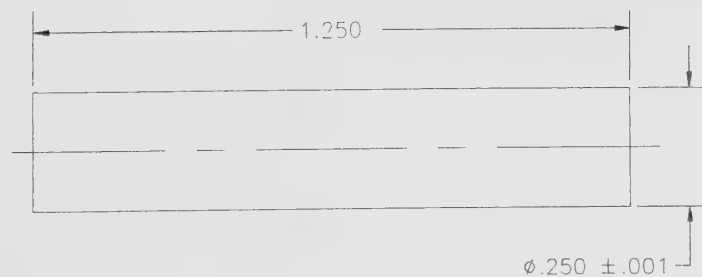
Part Name: Pump plunger

Part No.: 1DT1011

**⑪ PUMP PLUNGER**

Part Name: Drive pin

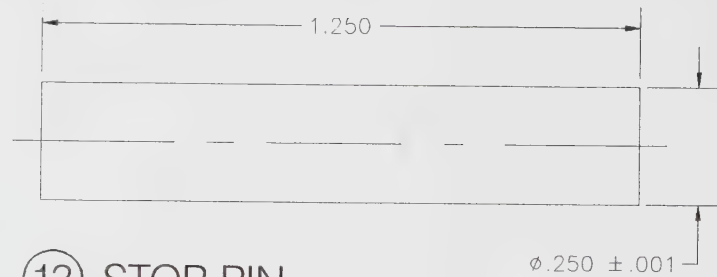
Part No.: 1DT1012

**⑫ DRIVE PIN**

PROBLEM 18.20 Hydraulic jack (Continued)

Part Name: Stop pin

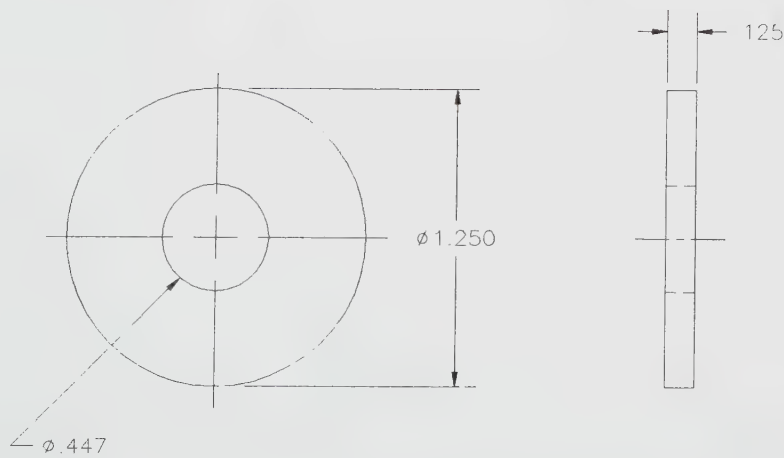
Part No.: 1DT1013



⑬ STOP PIN

Part Name: Piston washer

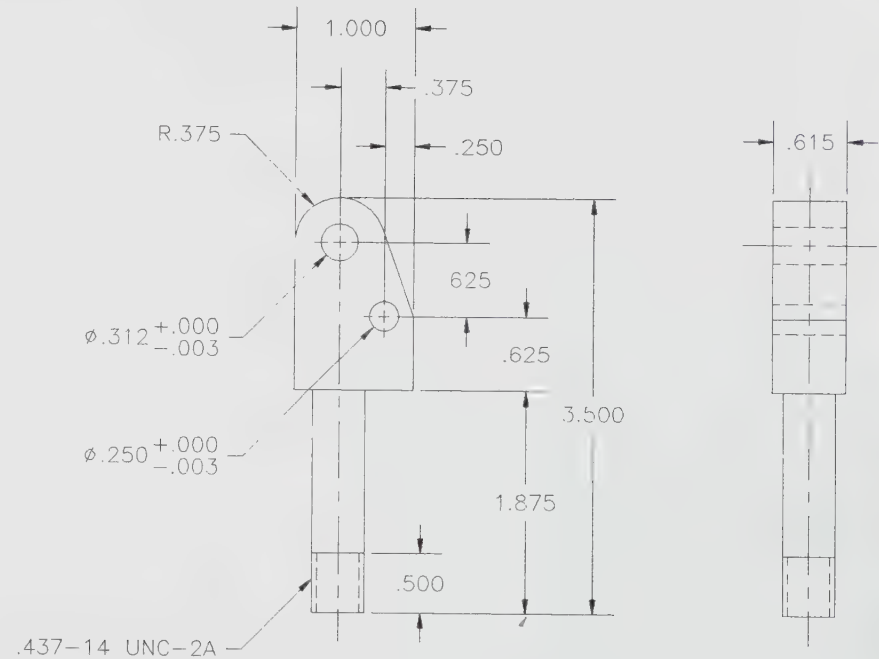
Part No.: 1DT1020



⑳ PISTON WASHER

Part Name: Pump pivot support

Part No.: 1DT1014

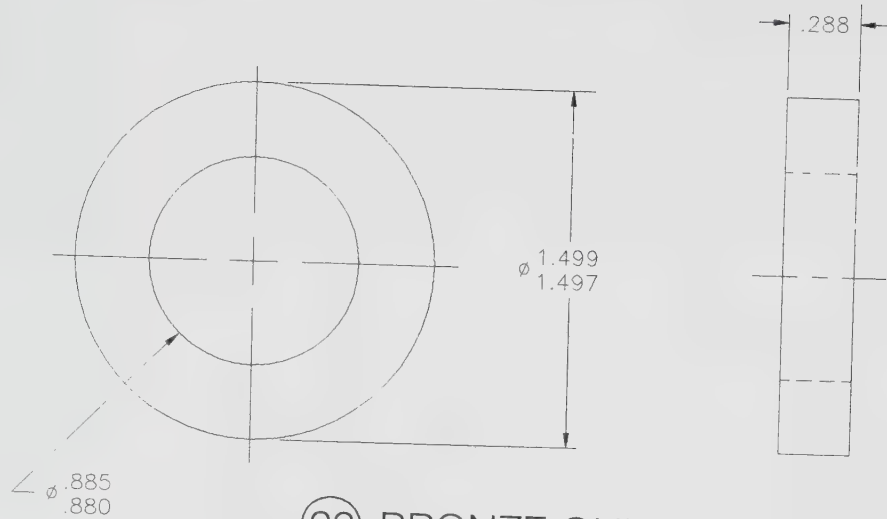


⑭ PUMP PIVOT SUPPORT

PROBLEM 18.20 Hydraulic jack (Continued)

Part Name: Bronze guide

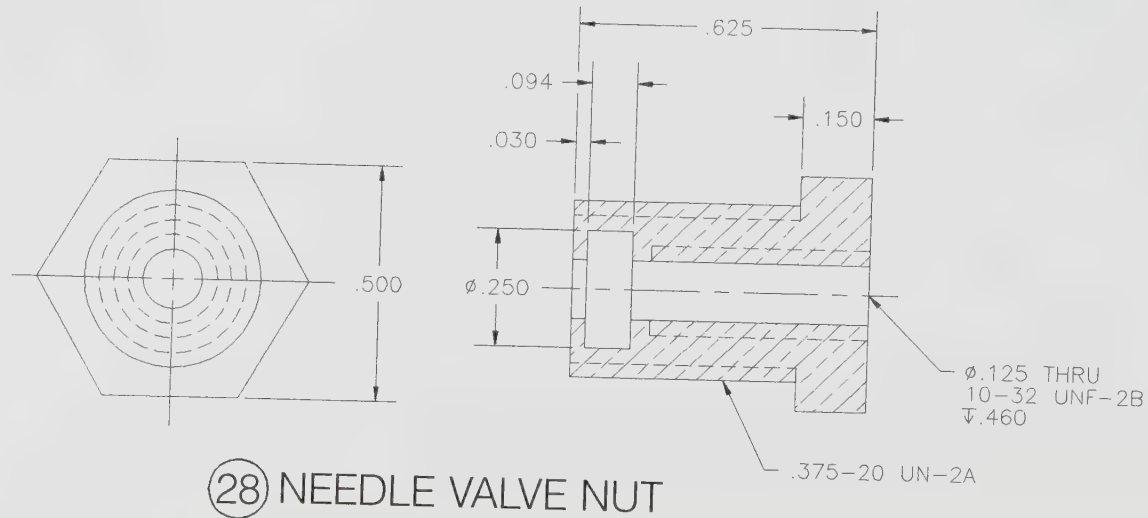
Part No.: 1DT1022



(22) BRONZE GUIDE

Part Name: Needle valve nut

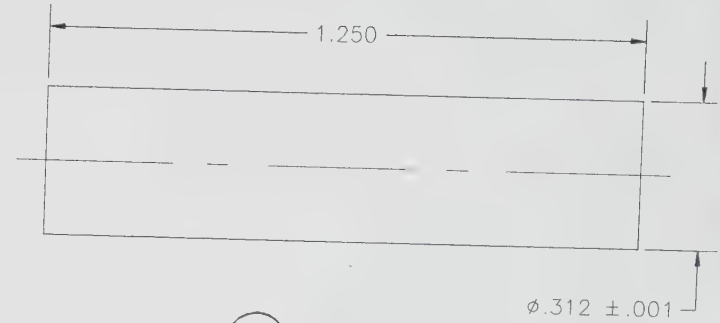
Part No.: 1DT1028



(28) NEEDLE VALVE NUT

Part Name: Pivot pin

Part No.: 1DT1023

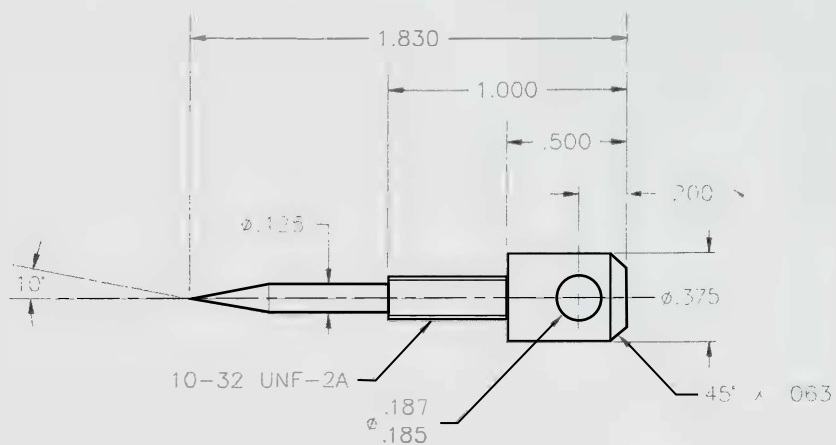


(23) PIVOT PIN

PROBLEM 18.20 Hydraulic jack (Continued)

Part Name: Needle valve

Part No.: 1DT1029



(29) NEEDLE VALVE

Part Name: Needle valve handle

Part No.: 1DT1030

2X SR.094



(30) NEEDLE VALVE HANDLE